

Inter-VLAN Routing Analysis Report

Cisco Packet Tracer Lab: inter routing vlan.pkt

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1. Objective

To analyse a Packet Tracer topology in which two departments are separated into different VLANs and subnets, and to explain how a router provides communication between them (inter-VLAN routing).

2. Network Topology

The screenshot below shows the topology as built in Packet Tracer (Logical view).

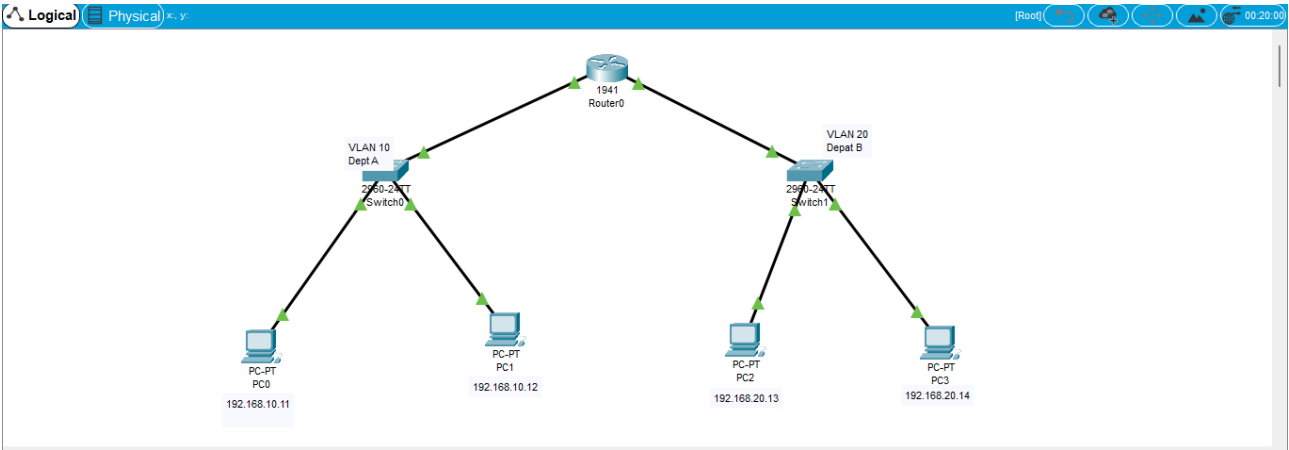


Figure 1: Inter-VLAN routing topology

The network has one Cisco 1941 router (Router0) connected to two Cisco 2960-24TT switches. Switch0 serves Department A (VLAN 10) with PC0 and PC1. Switch1 serves Department B (VLAN 20) with PC2 and PC3. The router sits between the two switches and is the only path between the departments.

3. Device and Addressing Table

Device	Type / Model	VLAN	Department	IP Address
Router0	Cisco 1941	Gateway	-	192.168.10.1 and 192.168.20.1 (assumed)
Switch0	Cisco 2960-24TT	10	Dept A	-
Switch1	Cisco 2960-24TT	20	Dept B	-
PC0	PC-PT	10	Dept A	192.168.10.11
PC1	PC-PT	10	Dept A	192.168.10.12
PC2	PC-PT	20	Dept B	192.168.20.13

Device	Type / Model	VLAN	Department	IP Address
PC3	PC-PT	20	Dept B	192.168.20.14

Note: the screenshot shows only the PC IP addresses. The subnet mask (/24) and the gateway addresses (.1 of each subnet) are the standard convention and are assumed here.

4. Analysis

4.1 Why VLANs?

A VLAN splits one physical network into separate broadcast domains. Dept A and Dept B hosts are in different VLANs, so broadcasts from one department never reach the other, which improves performance and security.

4.2 Subnet separation

Each VLAN uses its own IP subnet: VLAN 10 uses 192.168.10.0/24 and VLAN 20 uses 192.168.20.0/24. Hosts in different subnets cannot reach each other directly, even if they are cabled to the same switch, and need a Layer-3 device.

4.3 Role of the router

Router0 has one interface in each subnet, acting as the default gateway for both. When PC0 (192.168.10.11) pings PC2 (192.168.20.13), PC0 sees the destination is in another subnet, sends the frame to its gateway, the router routes the packet out the VLAN 20 interface, and Switch1 delivers it to PC2.

4.4 Traffic flows

Source	Destination	Path	Result
PC0	PC1	Switch0 only (same VLAN, Layer 2)	Reachable
PC2	PC3	Switch1 only (same VLAN, Layer 2)	Reachable
PC0 / PC1	PC2 / PC3	Switch0 - Router0 - Switch1	Reachable via router

4.5 Design observation

Because each switch carries a single VLAN and has its own link to the router, this design uses one physical router interface per VLAN. It works well for a small lab, but it uses one router port per VLAN. The more scalable method is **router-on-a-stick**: a single trunk link with sub-interfaces (for example g0/0.10 and g0/0.20) on a switch that carries both VLANs.

5. Sample Configuration

Switch0 (repeat for Switch1 with VLAN 20):

```
vlan 10
 name DeptA
interface range fa0/1 - 2
 switchport mode access
 switchport access vlan 10
```

Router0:

```
interface g0/0
  ip address 192.168.10.1 255.255.255.0
  no shutdown
interface g0/1
  ip address 192.168.20.1 255.255.255.0
  no shutdown
```

Interface and port numbers are examples; match them to the actual cabling in the .pkt file.

6. Verification Commands

Command	Purpose
ping <ip> (on PC)	Test same-VLAN and cross-VLAN connectivity
tracert <ip> (on PC)	Confirm the router is the intermediate hop
show vlan brief	Check VLAN-to-port assignment on switches
show ip interface brief	Check router interface status and IPs
show ip route	Confirm both subnets appear as connected routes

7. Conclusion

The topology isolates Dept A (VLAN 10) and Dept B (VLAN 20) at Layer 2 and uses Router0 to allow controlled communication at Layer 3. Hosts in the same VLAN communicate through their switch, while cross-VLAN traffic must pass through the router, demonstrating the core principle of inter-VLAN routing.